

REVIEW



Cryptic diversity begets challenges and opportunities in biodiversity research

Rui CHENG,¹ Arong LUO,¹ Michael ORR,^{1,2} Deyan GE,¹ Zhong'e HOU,¹ Yanhua QU,¹ Baocheng GUO,¹ Feng ZHANG,³ Zhongli SHA,⁴ Zhe ZHAO,¹ Mingqiang WANG,^{1,5} Xiaoyu SHI,¹ Hongxiang HAN,¹ Qingsong ZHOU,¹ Yuanning LI,⁶ Xingyue LIU,⁷ Chen SHAO,⁸ Aibing ZHANG,⁹ Xin ZHOU⁷ and Chaodong ZHU^{1,10,11}

¹Key Laboratory of Zoological Systematics and Evolution, Institute of Zoology, Chinese Academy of Sciences, Beijing, China,

²Entomologie, Staatliches Museum für Naturkunde Stuttgart, Stuttgart, Germany, ³College of Plant Protection, Nanjing Agricultural University, Nanjing, China, ⁴Institute of Oceanology, Chinese Academy of Sciences, Qingdao, China, ⁵Chengdu Institute of

Biology, Chinese Academy of Sciences, Chengdu, China, ⁶Institute of Oceanography, Shandong University, Qingdao, China,

⁷Department of Entomology, China Agricultural University, Beijing, China, ⁸College of Life Sciences, Shaanxi Normal University, Xi'an, China, ⁹College of Life Science, Capital Normal University, Beijing, China, ¹⁰State Key Laboratory of Integrated Pest

Management, Institute of Zoology, Chinese Academy of Sciences, Beijing, China and ¹¹College of Life Sciences/International College, University of Chinese Academy of Sciences, Beijing, China

Abstract

How many species of life are there on Earth? This is a question that we want to know but cannot yet answer. Some scholars speculate that the number of species may reach 2.2 billion when considering cryptic diversity and that each morphology-based insect species may contain an average of 3.1 cryptic species. With nearly two million described species, such high estimates of cryptic diversity would suggest that cryptic species are widespread. The development of molecular species delimitation has led to the discovery of a large number of cryptic species, and cryptic biodiversity has gradually entered our field of vision and attracted more attention. This paper introduces the concept of cryptic species, how they evolve, and methods by which they may be discovered and confirmed, and provides theoretical and methodological guidance for the study of hidden species. A workflow of how to confirm cryptic species is provided. In addition, the importance and reliability of multi-evidence-based integrated taxonomy are reaffirmed as a way to better standardize decision-making processes. Special focus on cryptic diversity and increased funding for taxonomy is needed to ensure that cryptic species in hyperdiverse groups are discoverable and described. An increased focus on cryptic species in the future will naturally arise as more difficult groups are studied, and thereby, we may finally better understand the rules governing the evolution and maintenance of cryptic biodiversity.

Key words: DNA barcoding, genome, integrative taxonomy, morphology, workflow

Correspondence: Chaodong Zhu, Key Laboratory of Zoological Systematics and Evolution, Institute of Zoology, Chinese Academy of Sciences, Beijing 100101, China.

Email: zhucd@ioz.ac.cn

Biodiversity deeply influences the lives of human beings (Díaz *et al.* 2006). Ecosystem services, the ways other species benefit humankind, are in many ways reliant on species biodiversity, as different species contribute in different ways. Yet, our understanding of global biodiversity remains incipient for many groups, such as arthropods (Stork 2018; Pfingstl *et al.* 2021).

How many species are there on Earth? In answering this question, the taxonomic challenges posed by cryptic/hidden species are non-negligible, resulting in variable estimates of species richness (Costello *et al.* 2013). One of the earliest such estimates was 0.4 million, made by Westwood (1833), followed much later by some estimates as high as 30 million or more (Erwin 1982), and more recent estimates have eventually settled on 8.7 million (Mora *et al.* 2011) and 5.5–8 million species (Stork 2018). Now, we face another wave of even higher estimates, reaching up to 2.2 billion (Li & Wiens 2022). However, as of 2021, roughly 2 million species have been formally described and named (Bánki *et al.* 2021), and half of them are insects. Even less than 1% of biodiversity may have been described, depending on which estimate of species richness is correct (Laurance 2013; Havermans 2016; Bánki *et al.* 2021; Li & Wiens 2022). As well, discovering cryptic species will prove essential to estimate the accurate number of species existing on Earth.

With the development of molecular biology technologies, multiple species that could not be discerned via morphology have been identified, and additional difficult species complexes also have become tractable. The concepts of “cryptic/hidden species” and “cryptic/hidden biodiversity” consequently came into being. Li and Wiens (2022) suggest that cryptic species may be widespread and common: that each morphology-based insect species may contain an average of 3.1 cryptic species. For hyperdiverse taxa such as insects, it is to be expected that discovery would be a multi-step process requiring such re-examination. The distribution of cryptic species diversity seems unlikely to be random, and this has important implications for speciation, evolution, biodiversity assessment, and conservation (Schönrogge *et al.* 2002; Lee & Ó Foighil 2004; Geml *et al.* 2006; Williams *et al.* 2012).

This review will introduce the concept and progress of their study, how they are thought to evolve, relevant discovery and research methods, and their significance. Our aim is to convince researchers to pay more attention to cryptic biodiversity and to more deeply understand evolutionary processes.

THE CONCEPT OF CRYPTIC SPECIES

The concept of cryptic species is considered an extension of that of species, and the prerequisite for defining a cryptic species entails defining “species.” In total, more than 30 species concepts were provided for respective purposes (Wilkins 2006, 2009; de Queiroz 2007; Orr *et al.* 2022). Among them, the biological species concept is the most well-known and widely used, stating that a species is a group of individuals that are reproductively isolated from other groups (Mayr 1963, 1982). The most commonly used concept of a cryptic species is also largely based on the biological species concept, defined as “species that are morphologically similar to known ones but have developed reproductive isolation” (Knowlton 1993; Adams 1998). In addition, some scientists agree the concept of cryptic species and the concept of species are incompatible based on points that “cryptic” itself derives from morphology and many species concepts do not include morphological evidence (e.g. Heethoff 2018; Pfingstl *et al.* 2021). Thus, we generally accept the theoretical underpinnings of the biological species concept as it pertains to the concept of cryptic speciation, which is the base of this review, although there are long-known major issues in its implementation that prevent its universal application (Ehrlich 1961; Sokal & Crovello 1970; Stamos 2003; de Queiroz 2007; Liu 2016; Orr *et al.* 2022). Specifically, in this review, we modified the concept of cryptic species as follows: “species owning subtle morphological traits but genetic divergence from the known species” to weaken “reproductive isolation.”

DESCRIPTION AND DEVELOPMENT OF CRYPTIC SPECIES

Interestingly, the idea of cryptic species has a long history. William Derham’s 1718 study of the bird genus *Phylloscopus* (Phylloscopidae) may have reported the very first potential cryptic species (Winker 2005). Until the early 2000s, there were few studies about cryptic species, however (Fig. 1). After that, a series of researchers proposed “DNA barcoding” and explicitly extended it to the study of cryptic species (Tautz *et al.* 2002; Hebert *et al.* 2003a,b; Blaxter 2004; Xu & Che 2019). This was then a critical time node in the development history of the cryptic species research, being supported by Web of Science queries. Using the keywords “cryptic/hidden species” etc. to search, we found that most (around 95%) of the related research literature was published after this period, beginning to grow rapidly around 2006. Usually, the discovery

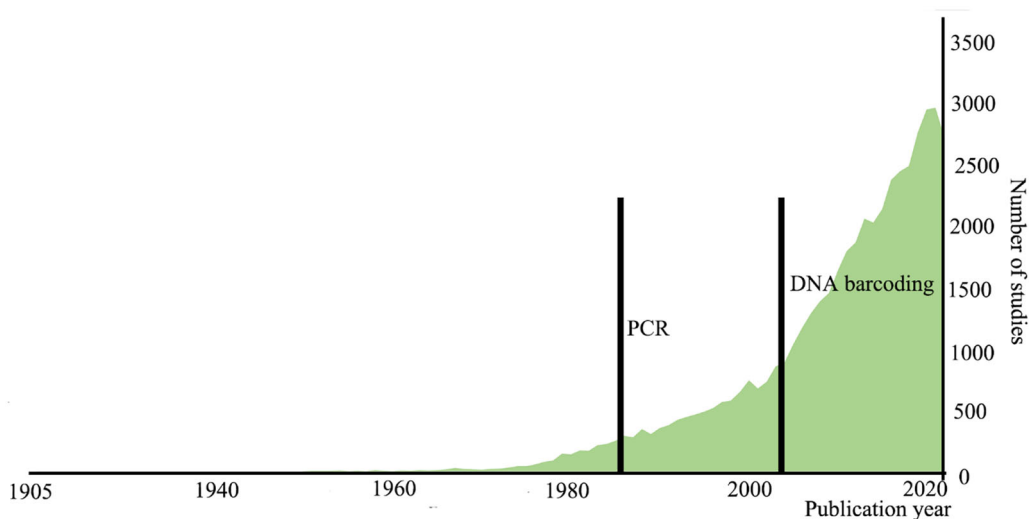


Figure 1 Numbers of published literature from the Web of Knowledge about cryptic/hidden species per year.

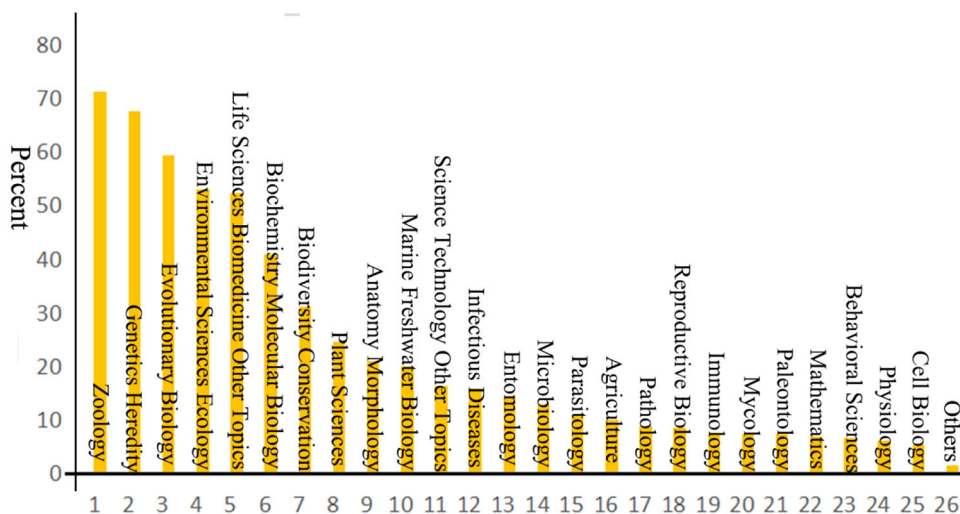


Figure 2 The research fields and percentages of published literature related to cryptic/hidden species.

and confirmation of cryptic species are byproducts of other related biological fields, mainly including zoology, genetic heredity, and evolutionary biology (Fig. 2). Currently, the study of cryptic species has attracted the attention of more biologists, opening a new window for understanding the mechanisms behind speciation and injecting new ideas, methods, and rigor into biodiversity science.

Among all the groups of organisms, one significant bias is the preponderance of animals in published papers (Fig. 3), and almost all animal taxa include cryptic species (Fig. 4). Summarizing these papers, several animal groups were thought to own relatively more common cryptic species, for example, protozoans, arthropods, and frogs.

Protozoans, as a paraphyletic conglomeration of single-celled animals, are widely present taxa with many cryptic species due to the lack of clear defining morphological features and the difficulty of collection (Gong *et al.* 2013; Fan *et al.* 2021; Liu *et al.* 2022). Another fascinating group of cryptic diversity is arthropods, many groups of which can be considered as a “taxonomist’s nightmare” (Bickford *et al.* 2007; Pfingstl *et al.* 2021). Among them, there are groups where the majority of species are likely to be undescribed and difficult to tell apart (Chimeno *et al.* 2022). There are many possible ways that species might be cryptic, based on characters or habits in addition to typical morphological differentiation. Frogs and some

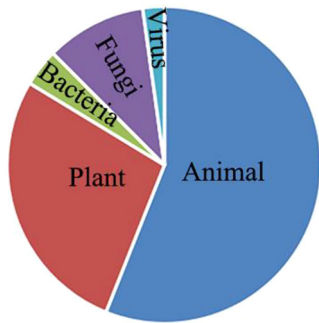


Figure 3 The proportion of five main biological groups (animal, plant, bacteria, fungi, and virus) related to cryptic/hidden species.

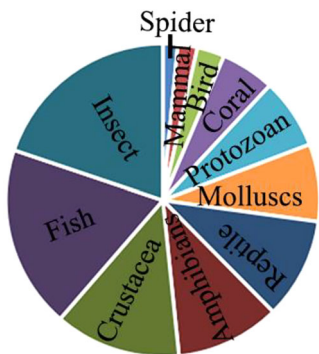


Figure 4 The proportion of 11 main animal groups related to cryptic/hidden species.

other groups, for example, likely exhibit many cryptic species based on mate selection systems relying on non-morphological features, such as songs (Narins 1983).

HYPOTHESES OF CRYPTIC SPECIATION

In this review, we mainly focus on morphology-conserved cryptic species and how they are formed, which can be best understood as a “special speciation mechanism.” Here, we summarize four non-mutually exclusive evolutionary mechanisms that have been hypothesized to lead to possibly non-random cryptic species (Bickford *et al.* 2007; Kress *et al.* 2015; Struck & Cerca 2019; Pfinstl *et al.* 2021).

Recent divergence

The simplest explanation for cryptic speciation is that they have simply not had time to develop differences in all forms, sometimes called “species on the road to differentiation” (Liu 2016). Species identifications that rely

solely on morphology may overlook such recently diverged species (Holland *et al.* 2004), especially if performed by a non-expert. This idea is based on the notion that morphological traits should follow a neutral evolutionary model (i.e. that morphological differences accumulate linearly over time; Felsenstein 1985). It is posited that in the early stages of speciation, selection pressure may act on physiological, reproductive, behavioral, niche, and other traits, and the effect occurs at a relatively late stage for morphological characteristics (Damm *et al.* 2010; Cheng *et al.* 2016; Derycke *et al.* 2016). There are many related examples. In the case of the genus *Rattus* (Muridae), which has undergone a recent species radiation (Middle Pleistocene), morphological differences between clades are very small (Rowe *et al.* 2011). Given the commonness of this phenomenon, much research has gone into alternative lines of evidence.

Molecular evidence in many cases may show significant differentiation in the absence of distinctive morphological characters, resulting in splitting at various levels, such as for insects (Chesters *et al.* 2012; Jiang *et al.* 2014; Cheng *et al.* 2016; Zhou *et al.* 2017), mammals (Andersen & Light 2012), and protozoans (Shao *et al.* 2019). For example, the formation of a large number of alpine lakes in the Qinghai–Tibet Plateau and Alps during recent interglacial cycling drove the differentiation of the amphipod *Gammarus lacustris*; the morphology of each clade was similar, but the genetic distances between clades ranged from 2% to 8% for mitochondrial DNA (Hou *et al.* 2022).

Some additional special evolutionary phenomena may occur in the early stages of speciation that can profoundly influence morphological evolution and complicate species delineation, such as budding speciation (a consequence of incomplete lineage sorting; Harrison 1991; Funk & Omland 2003; Kruckenhauser *et al.* 2014; Cheng *et al.* 2017) and hybridization (Mallet 2005; Weigand *et al.* 2017). For example, a study of one Tibetan Plateau frog species (*Scutiger mamatus*) found that both budding speciation and incomplete lineage sorting prompted the formation of a cryptic species (Chen *et al.* 2009). During hybridization, intermediate morphological populations with both parents’ genetic traits in a hybrid zone may form cryptic species (Barton & Hewitt 1985; Harrison 1993; Wu *et al.* 2011; Gong *et al.* 2013; Wang *et al.* 2014).

Morphological stasis

Studies have found that the process of speciation is not necessarily accompanied by morphological changes. In addition to more straightforward biological reasons, such as low mutation rates, or developmental and

physiological limitations, selection pressures are important potential causes of morphological stasis (Grundt *et al.* 2006; Struck & Cerca 2019; Pfingstl *et al.* 2021). In some environments, selection pressures mainly act on traits such as behavior and physiology, while morphological traits that are not closely related to survival evolve very slowly, and thereby do not necessarily change with speciation. In some parasitic insects, for example, selection pressures may primarily act on traits related to parasitism (Schönrogge *et al.* 2002), such as the ability to detect hosts or counteract their defenses, while morphological traits unrelated to parasitism may experience morphological stagnation, as changes in them might reduce their parasitic efficiency (Chen 1946; Kaleka *et al.* 2020).

Morphological stasis is especially common in extreme environments, where organisms do not have many options for adaptation when dealing with harsh living conditions, and physiology and related morphological traits are strongly constrained (Rothschild & Mancinelli 2001; Nevo 2001). This explains why large numbers of cryptic species are found in habitats such as Arctic tundra, underwater karst landforms, caves, and deep-sea environments (Vrijenhoek *et al.* 1994; Witt *et al.* 2006; Lefébure *et al.* 2006; Hou & Li 2010). Cave spiders, for instance, show little morphological variation, due to adaptations such as eye degradation, light body color, and slender legs, and there are consequently many cryptic species only discovered with extensive examination (Ba & Lai 2009). Indeed, species complexes are fairly common in cave ecosystems (Trontelj 2009; Derkarabetian *et al.* 2010; Hedin 2015; Delić *et al.* 2017).

Traits may also be strongly conserved by behaviors, such as the use of narrow cavities as nests by the bee family Megachilidae: Nearly all species that make nests carry their pollen on their underside rather than on their legs (Michener 2007), assumedly to avoid pollen rubbing off their legs in these confining environments. Many of the traits that are needed to survive in such specific environments are interlinked with ancillary traits, ultimately inhibiting changes in them that might impact the functionality of the traits needed to survive. Under such circumstances, it is no surprise that morphological stasis may occur.

Nonvisual signals

Cryptic species are commonly found in organisms that distinguish each other by nonvisual signals (Mayr 1963; Ruxton 2009; Cooke *et al.* 2012). Nonvisual mating signals (i.e. acoustic signals, chemical mating signals) have been used to identify insect relatives or cryptic species

(Henry 1994; Marshall & Cooley 2000; Angulo & Reichle 2008; Martinet *et al.* 2019).

Acoustic signals, such as chirping, are commonly used to distinguish between birds (Cicero 1996; Lei & Payne 2002; Yang & Lei 2008), frogs (Narins 1983), insects (Henry 1994; Henry & Wells 2010), and bats (Jones & Barlow 2004). Perhaps the most famous examples come from birds, where songs play a central role in reproductive isolation between species (Yang & Lei 2008; Lei *et al.* 2003). Based on differences in songs, several cryptic species were identified, such as the Himalayan cuckoo (*Cuculus saturates*) and Horsfield's cuckoo (*C. saturatus* and *C. horsfieldi*) (Lei & Payne 2002). Another important case is among insect songs, which can help entomologists discover cryptic species. For example, two cryptic sibling species of *Chrysoperla* green lacewings were supported by acoustic niche partitioning (Henry *et al.* 2003; Henry & Wells 2010).

Convergent evolution and parallel evolution

Although the concepts of convergent and parallel evolution are distinguishing, they can lead to similar morphological characteristics, which are often difficult to distinguish (Belyaeva & Taylor 2009; Pearce 2011; Struck & Cerca 2019). Unlike the recent divergence of speciation hypothesis, where cryptic species are closely related sisters, the convergent evolution hypothesis invokes distantly related species that evolve similar morphological characteristics independently (Holland *et al.* 2004; Struck & Cerca 2019), such as the evolution of wings in birds, bats, and insects. Similarly, Moen *et al.* (2013) suggest convergent evolution led to a striking similarity in the morphology of frogs. The convergent evolution hypothesis states that some cryptic species have evolved independently from morphologically distinct ancestors (Swift *et al.* 2016; Struck *et al.* 2018). In the early stages of differentiation, such species exhibit similar rates of morphological differentiation. However, at some point in time, some species began to converge in their morphological trajectories, forming cryptic species.

In contrast, the parallel evolution hypothesis involves related species, where descendants of a common ancestor adapt to similar living environments and independently develop similar morphology (Holland *et al.* 2004; Pearce 2011). A typical example of this hypothesis is that Australian marsupials have similar counterparts to true mammals in Eurasia (Weisbecker & Archer 2008). In an additional example, Yamamoto and Sota (2007) found that wingless females of the family Geometridae evolved independently in three subfamilies, to adapt to climate cooling events in the Oligocene and Early Pleistocene.

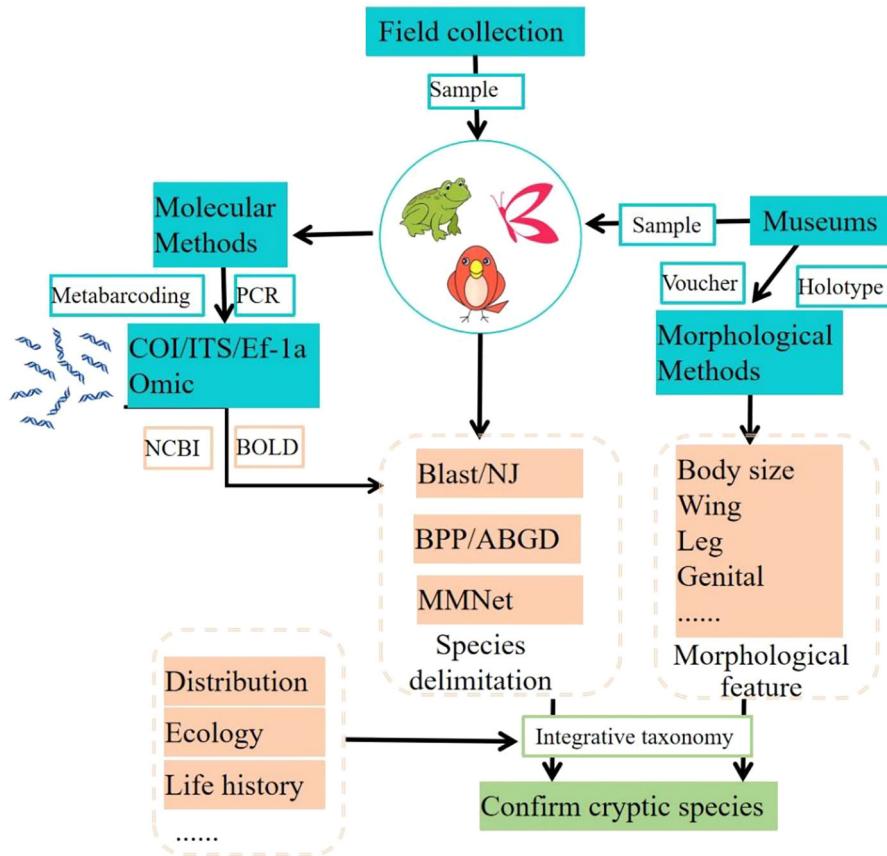


Figure 5 Workflow of confirming cryptic species. COI, cytochrome c oxidase subunit I.

DISCOVERY AND CONFIRMATION OF CRYPTIC SPECIES

Technological advances in molecular biology have been central to the study of cryptic species, and a workflow of how to confirm cryptic species derived from molecular evidence is given in Fig. 5. This includes work based on a wide variety of data types and densities, including single mitochondrial genes (COIs), multiple mitochondrial genes, whole mitochondrial genomes, reduced-representation genomes, and genome-wide data (e.g. SNPs). At the same time, morphologists have made great progress in the last several decades, transitioning from classical anatomical morphological characteristics to more detailed morphological studies based on geometric morphometrics, electron microscopy scanning, and micro-CT.

Integrative taxonomy, combining molecular, morphological, ecological, and other sources of evidence, has become key to the synthesis and use of such disparate data

types, enabling more effective means of exploring biodiversity and discovering and delimiting cryptic species (Dayrat 2005; Padial *et al.* 2010; Hartop *et al.* 2022; Orr *et al.* 2022). Kergoat *et al.* (2015) combined ecological, morphological, and molecular data to discover five cryptic species within it, providing a sounder theoretical basis for the precise control of pest species in this genus. In another example, it was long thought that there were only 10 species of the common Chinese white-bellied rat genus *Niviventer*. However, by integrating external and cranio-dental measurements with multi-locus molecular data, Ge *et al.* (2018, 2021) delimited 18 species in China alone. However, the ways in which these data are used, including how different lines of evidence are weighed, can have a substantial impact on the results of such biodiversity investigations. It is therefore vital to understand both the data and how they can reasonably be used.

Here, we focus our methodological review primarily on molecular data and how these may be used to explore cryptic species and cryptic biodiversity.

Single locus or multiple loci DNA barcoding

The most revolutionary development in the study of cryptic species has been the widespread adoption of molecular methods spurred by DNA barcoding and similar initiatives for a system of DNA taxonomy (Hebert *et al.* 2003a; Tautz *et al.* 2003). DNA barcoding technology was first proposed in 2003 to rapidly discover cryptic species, delimit species, and reveal their genetic structure through a gene fragment that is highly variable (quickly evolving) and relatively easy to amplify for many animal taxa (Hebert *et al.* 2003a,b, 2004; Xiao *et al.* 2004; Peng *et al.* 2008; Liu *et al.* 2010). The agreed-upon animal marker, mitochondrial cytochrome c oxidase subunit I (COI), has been widely used in invertebrates, birds, fish, and beyond (Hebert *et al.* 2003a,b, 2004; Costa *et al.* 2007; Liu *et al.* 2010; Shen *et al.* 2019; Xiong *et al.* 2022). As an independent line of evidence, COI and similar markers, such as 16S rRNA for bacteria, provide an invaluable link between life stages and dimorphic sexes, while also making it easier to account for complexities such as phenotypic plasticity.

Driven by the International Barcode of Life (iBOL) project and immense community inputs, the BOLD website has so far collected 15 466 718 animal barcode sequences from 247 000 species as of February 7, 2023, with arthropods (13 899 279) representing 89.86%. Based on barcode technology, many cryptic species have been found in almost all animal groups (Hebert *et al.* 2004; Costa *et al.* 2007; Wang *et al.* 2020; Yu & Wang 2021). More recently, the approach has been extended to other braconid wasps to hasten biodiversity discovery and species description of hyperdiverse groups (Sharkey *et al.* 2021). Another benefit of this approach is, ideally, an inherent standardization of the species delimitation process, such that theoretically, if applied correctly, all species would be recognized in the same, replicable manner within the group (Orr *et al.* 2022).

In addition, technological advances have made larger-scale data and analyses feasible over the last 20 years. Except traditional acquisition of DNA sequences based on PCR amplification of a single sample, eDNA metabarcoding technology uses high-throughput approaches to obtain batches of amplicon sequences in mixed samples (Ruppert *et al.* 2019). For mixed insect samples from light traps, pitfall traps, malaise traps etc., metabarcodes have been repeatedly leveraged (Zhou *et al.* 2013; Tang *et al.* 2015), and in some cases, even just residual ethanol used to preserve such samples may be sequenced without damaging specimens (Ritter *et al.* 2019; Mata *et al.* 2021; Zenker *et al.* 2020). For example, Huang *et al.* (2022) found that 95% of the species of the dipteran insect com-

munity from Tianmu Mountain based on metabarcodes were unidentified or undescribed species.

There have been many critiques of this approach, however, as minimalist treatments may make it harder to do future taxonomic work on these groups (Fernandez-Triana 2022). Almost complete reliance on molecular data in tropical countries could, for instance, make it much harder for local researchers with limited access to funds and facilities for molecular analysis to identify local species. Another primary issue is the questionable practice of relying solely on a single gene fragment for species definition (Meier *et al.* 2022), which is especially problematic when the data are not fully public and analyses are not replicable. For example, across varied taxa such as collembolans, spiders etc., the results of species delimitation using COI+ITS2 are better than when using COI alone (Agnarsson *et al.* 2007).

One suggested fix for this is to use multiple markers, especially to add nuclear markers (Fišer *et al.* 2018; Després 2019), because maternal mitochondrial genes may prove ineffective in discovering cryptic species undergoing a special evolutionary process (Funk & Omland 2003). For example, the moth cryptic species *Limbatoclamys pararosthorni* and its sister species *L. rosthorni* were supported by the nuclear gene (Ef-1a), not by COI (Han *et al.* 2007; Jiang *et al.* 2021). Multiple barcoding markers can find the traits of pseudo-cryptic species and try to avoid several complex evolutionary processes such as mito-nuclear discordance and paraphyly. Mito-nuclear discordance is widespread across certain taxa like copepods (Barreto *et al.* 2018) and insects (Jiang *et al.* 2021), and the possible reasons include gene introgression, incomplete lineage sorting, and *Wolbachia* infection within arthropods. For example, Weigand *et al.* (2017) confirmed two cryptic species with mito-nuclear discordance induced by gene introgression.

Omics-based research on cryptic species

Although many studies of cryptic species are based on DNA barcoding, when facing some complex evolutionary processes, genomic data provide greater promise for the resolution of species. Janzen *et al.* (2017) suggested that COIs are suited to expose potential cryptic species, whereas nuclear genomes are used to distinguish/confirm cryptic species. Thankfully, commonly cryptic species identified by single mtDNA gene are also supported by omics data (Janzen *et al.* 2017; de Moya *et al.* 2019).

However, sometimes omics data rebut cryptic clades identified by single mtDNA genes (Cong *et al.* 2017; Hinojosa *et al.* 2019; Hupalo *et al.* 2023). Many potential causes have been discussed previously (introgression

etc.). Hinojosa *et al.* (2019) suggested homogeneity of the nuclear genome via long periods of geographic isolation also can deny the validity of cryptic species based on COI. More cases of mitochondrial genome introgression have gradually been discovered in different animal groups, demonstrating that some molecular cryptic species for barcodes might have been products of introgression (Cong *et al.* 2017; Ge *et al.* 2022). For example, several well-studied and emblematic butterfly groups owning striking mtDNA introgression, such as *Erynnis* (family Hesperidae), genome data revealed greater reliability than COI (Zakharov *et al.* 2009; Cong *et al.* 2017).

Complicating matters, introgression may have even occurred historically with now-extinct species, known as “ghost introgression” (Ottenburghs 2020). This phenomenon can occur when species are undergoing sudden range or habitat expansions, bringing them into contact with other closely related species. With the introduction of new genetic material and its potential fixation, it would then be much harder or even impossible to use that hypothetical marker to accurately delimit species. In some cases, these species might also be morphologically cryptic, in which case genomic methods become truly invaluable.

Facing these complex evolutionary processes, genome scanning is a useful method to discover potential cryptic species by locating key regions of genetic differentiation between candidate species, such as the genomic island of speciation and introgression segments (Utsunomiya *et al.* 2013; Steane *et al.* 2015; Chattopadhyay *et al.* 2016; Després 2019). In the future, pointing omic islands of speciation, containing a series of linked sites related to reproductive isolation, may be one important method to distinguish cryptic species.

APPLIED IMPORTANCE OF CRYPTIC SPECIES RESEARCH

The prevalence of cryptic species has clear and substantial implications beyond the bounds of systematics and evolutionary studies. Nowadays, cryptic species have profound effects on many fields, including biodiversity, conservation, pest management, and so on. Here, we will discuss four such distinctive and important cases.

Important advancements for morphological taxonomy

Classical taxonomy based on morphological features is highly reliant on taxonomic knowledge and verified museum specimens (Orr *et al.* 2020; Hong *et al.* 2022;

Zhu *et al.* 2022). Taxonomic knowledge requires substantial training and experience and their practices are often relatively more subjective in comparison across groups; molecular analyses and other lines of evidence spurred in their use by cryptic species discovery can enable more researchers to freely explore species delimitation while offering additional methods for standardization (such as through novel species delimitation algorithms designed for molecular data). Meanwhile, verified reference material, especially type specimens, can be difficult to check for taxonomists across the Global South, as many types of verified specimens are held in Western institutions. Thankfully, barcoding codes are typically more open-source, held on repositories such as BOLD or NCBI.

Molecular data are also invaluable for making associations that might be otherwise unclear based on morphology alone. This is especially important for associating different sexes, larval and adult stages, or those that differ in appearance by season. For example, potential cryptic species of insects and echinoids have been revealed by larval barcoding (Suh *et al.* 2019; Collin *et al.* 2020).

A critical role for biodiversity research

The assessment of species diversity is inseparable from the number of species present in an area, and this includes cryptic species. It is plausible that the discovery of cryptic species can significantly increase the diversity of communities, that is, α and β diversity (Vodá *et al.* 2015a; Darwell & Cook 2017; Mayr *et al.* 2021). Immense hidden biodiversity is also expected in some specific areas or naturally fragmented habitats, such as oceans, rivers, islands, and caves (Barber *et al.* 2006; Witt *et al.* 2006; Liu *et al.* 2011; Chen *et al.* 2015; Suh *et al.* 2019; Wang *et al.* 2020). For example, in recent years, the diversity of cave ecosystems has caused great concerns due to the discovery of plenty of hidden spider species (Ba & Lai 2009; Hedin 2015; Trontelj 2009; Delić *et al.* 2017). Zhang and Li (2014) discovered 15 undescribed cryptic species from the spider *Pinelema cucurbitina* in southern China karst cave, but even some vertebrates such as bats are likely to have many cryptic species (Chornelia *et al.* 2022).

Targeted protection of rare and beneficial species

Biodiversity conservation is an increasingly important goal for governmental or non-governmental organizations, as well as the general public, and the understanding of cryptic species can dramatically influence how we go about conservation. Nowadays, many biologists have realized the importance of cryptic species for conservation and acknowledged that cryptic species may

be more vulnerable to threats than previously thought (Nair *et al.* 2012; Niemiller *et al.* 2013; Delić *et al.* 2017). In recent years, many cryptic species were discovered within flagship protected species, including in the International Union for Conservation of Nature Red List. Examples include giraffes (Fennessy *et al.* 2016), finless porpoises (Zhou *et al.* 2018), and giant salamanders (Yan *et al.* 2018; Chai *et al.* 2022). Because of the differences in their distributions, population health, and degrees of endangerment among such related species, these cryptic species may decrease the effectiveness of management if not accounted for and increase unnecessary investments. Thus, timely and effective taxonomic efforts are needed to ensure that we are protecting the right species.

Targeted identification and control of pest and invasive species

The accurate identification of pests is crucial for effective pest management strategies. The existence of cryptic species in pest species can decrease management efficiency and increase unnecessary investments (Simmons & Scheffer 2004; Hendrichs *et al.* 2015; Andrews *et al.* 2020), and detection of cryptic species in economically important taxon should consequently be cautious. For example, the important agricultural pest thrips *Thrips palmi* likely contains three cryptic species that are partially sympatric (Tyagi *et al.* 2017), such that variance in behavior or susceptibility to pesticides might differ between them and require separate management strategies. One important example of cryptic species within invasive species is that of whiteflies. Whiteflies are disastrous pests prone to widespread spread and often require specialized management, but they were considered an untenable species complex until Dinsdale *et al.* (2010) found at least 24 cryptic species based on molecular data (Luo *et al.* 2002).

UNSOVLED BUT INTERESTING PHENOMENON ABOUT CRYPTIC SPECIES

Although taxonomic challenges involved in cryptic biodiversity have been recognized for hundreds of years, forming the foundation of challenges in telling species apart by morphology, many fascinating unresolved questions remain (Bickford *et al.* 2007; Tattersall 2007; Korshunova *et al.* 2017).

Are cryptic species evenly distributed among different taxa and various geographic regions? Through exploration of the published taxonomic literature, there has always been an illusion that cryptic species are more common in the tropics than in temperate areas, in insects than vertebrates (Bickford *et al.* 2007; Dyer *et al.* 2007). This idea is related to the expected high species diversity of tropical areas, as two-thirds of species might exist in such areas (Willig *et al.* 2003; Pekár 2014). However, inconsistently, most of the published studies focused on cryptic species of temperate organisms (Bickford *et al.* 2007). This discrepancy is likely a result of efforts made in temperate areas of the Global North, where there has historically been more taxonomic work (Janzen *et al.* 2005; Dyer *et al.* 2007). Among different taxa, cryptic diversity is thought to be especially common within insects (Bickford *et al.* 2007; Murray *et al.* 2008) because of their complex life history, limited and small morphological features etc. (Van Campenhout *et al.* 2014). Pfenninger and Schwenk (2007) suggested that cryptic animal species are evenly distributed among various regions and taxa based on one detailed taxonomic list and distribution information of cryptic species reports, but this requires further exploration.

Does cryptic speciation prefer allopatric, parapatric, or sympatric speciation? Cryptic speciation has been reported for all these types of speciation, for example, allopatry (Amor *et al.* 2014; Fišer *et al.* 2018), parapatry (Dennis & Hellberg 2010), and sympatry (Scriven *et al.* 2016). However, because of biases in the interests of researchers, where they work, and what they work on, confirming the relative prevalence of cryptic speciation across these forms of speciation is likely to prove even more challenging than for proportions in groups or areas, as each such case requires greater evidence, including for the specific mode of speciation.

Can cryptic species occur alongside known species? How often does this happen? For cryptic species, an enduring question of active research is whether such species can evolve together and co-occur (Zhang *et al.* 2004). One point supported no-occurrence between cryptic species based on species interactions (Vodá *et al.* 2015a,b; Garcia-Robledo *et al.* 2016; Darwell & Cook 2017). Conversely, others believe that hidden species can and do co-occur, either because the rate of competitive exclusion is low and slow-acting, or because they are not strictly ecologically identical (Scriven *et al.* 2016; Johnston *et al.* 2022) that they differ enough to avoid undue impacts. Key to answering this question will be looking for additional lines of evidence by which these species might distinguish themselves.

CONCLUSION AND OUTLOOK

Cryptic species have now been discovered across much of the Tree of Life, from every corner of the globe (citations from above). Going forward, we need to better account for cryptic species in biodiversity research and beyond, and we are constantly developing better methods and frameworks for these purposes. The fields of pest management and invasion biology present especially important arenas for future work. From these and other studies, we are gradually building a better understanding of how both species and cryptic species evolve, a fundamental question in biology. A growing number of studies confirm the importance and reliability of integrative taxonomy using multiple lines of evidence, but taxonomic researchers face more challenges in hiring and funding than ever before, even as awareness of the biodiversity crisis grows. Therefore, in the future, we hope that more scholars will pay attention to cryptic species, participate in this research, integrate various methods, and strive to explore the mechanism and evolutionary mechanisms underlying cryptic species. Cryptic species are ultimately a crux in the current biodiversity conservation framework, and only by understanding them can we effectively know and protect species.

ACKNOWLEDGMENTS

We are thankful to Prof. Keping Ma, Prof. Yibo Hu, Prof. Yue-Hong Yan, Yunxia Luan, and Guannan Wen for their strong support and valuable comments on this paper. This study was supported by the Ningxia Hui Autonomous Region Agricultural Science and Technology Independent Innovation Fund (NGSB20211405), Key Laboratory of the Zoological Systematics and Evolution of the Chinese Academy of Sciences (2008DP173354), and the Survey of Wildlife Resources in Key Areas of Tibet (ZL202203601).

CONFLICT OF INTEREST

The authors declare no competing interests.

REFERENCES

- Adams BJ (1998). Species concepts and the evolutionary paradigm in modern nematology. *Journal of Nematology* **30**, 1–21.
- Agnarsson I, Kuntner M (2007). Taxonomy in a changing world: Seeking solutions for a science in crisis. *Systematic Biology* **56**, 531–39.
- Amor MD, Norman MD, Cameron HE, Strugnell JM (2014). Allopatric speciation within a cryptic species complex of Australasian octopuses. *PLoS ONE* **9**, e98982.
- Andersen JJ, Light JE (2012). Phylogeography and subspecies revision of the hispid pocket mouse *Chaetodipus hispidus* (Rodentia: Heteromyidae). *Journal of Mammalogy* **93**, 1195–215.
- Andrews KR, Gerritsen A, Rashed A *et al.* (2020). Wireworm (Coleoptera: Elateridae) genomic analysis reveals putative cryptic species, population structure, and adaptation to pest control. *Communications Biology* **3**, 489.
- Angulo A, Reichle S (2008). Acoustic signals, species diagnosis, and species concepts: the case of a new cryptic species of *Leptodactylus* (Amphibia, Anura, Leptodactylidae) from the Chapare region, Bolivia. *Zoological Journal of the Linnean Society* **152**, 59–77.
- Ba JW, Lai DY (2009). Diversity of the cave-dwelling spiders and their adaptation to the cave environment in China. *Acta Zootaxonomica Sinica* **34**, 98–105. (In Chinese with English abstract.)
- Barber PH, Erdmann MV, Palumbi SR (2006). Comparative phylogeography of three codistributed stomatopods: origins and timing of regional lineage diversification in the Coral Triangle. *Evolution* **60**, 1825–39.
- Barreto FS, Watson ET, Lima TG *et al.* (2018). Genomic signatures of mitonuclear coevolution across populations of *Tigriopus californicus*. *Nature Ecology & Evolution* **2**, 1250–57.
- Barton NH, Hewitt GM (1985). Analysis of hybrid zones. *Annual Review of Ecology and Systematics* **16**, 113–48.
- Belyaeva M, Taylor DJ (2009). Cryptic species within the *Chydorus sphaericus* species complex (Crustacea: Cladocera) revealed by molecular markers and sexual stage morphology. *Molecular Phylogenetics and Evolution* **50**, 534–46.
- Bickford D, Lohman DJ, Sodhi NS *et al.* (2007). Cryptic species as a window on diversity and conservation. *Trends in Ecology Evolution* **22**, 148–55.
- Blaxter ML (2004). The promise of a DNA taxonomy. *Philosophical Transactions of the Royal Society B: Biological Sciences* **359**, 669–79.
- Bánki O, Roskov Y, Vandepitte L *et al.* (2021). Catalogue of Life Checklist (Version 2021-08-25). Catalogue of Life. Available from URL: <http://www.catalogueoflife.org>
- Chai J, Lu CQ, Yi MR *et al.* (2022). Discovery of a wild, genetically pure Chinese giant salamander creates new

- conservation opportunities. *Zoological Research* **43**, 469.
- Chattopadhyay B, Garg KM, Kumar AK *et al.* (2016). Genome-wide data reveal cryptic diversity and genetic introgression in an Oriental cynopterine fruit bat radiation. *BMC Evolutionary Biology* **16**, 41.
- Chen SH (1946). Evolution of the insect larva. *Transactions of the Royal Entomological Society of London* **97**, 381–404.
- Chen W, Bi K, Fu J (2009). Frequent mitochondrial gene introgression among high elevation Tibet megophryid frogs revealed by conflicting gene genealogies. *Molecular Ecology* **18**, 2856–76.
- Chen W, Ma X, Shen Y, Mao Y, He S (2015). The fish diversity in the upper reaches of the Salween River, Nujiang River, revealed by DNA barcoding. *Scientific Reports* **5**, 17437.
- Cheng R, Jiang N, Xue D, Li X, Ban X, Han H (2016). The evolutionary history of *Biston suppressaria* (Guenée) (Lepidoptera: Geometridae) related to complex topography and geological history. *Systematic Entomology* **41**, 732–43.
- Cheng R, Xue D, Galsworthy A, Han H (2017). Complete mitochondrial genomes throw light on budding speciation in three *Biston* species (Lepidoptera, Geometridae). *Zoologica Scripta* **46**, 73–84.
- Chesters D, Wang Y, Yu F *et al.* (2012). The integrative taxonomic approach reveals host specific species in an Encyrtid parasitoid species complex. *PLoS ONE* **7**, e37655.
- Chimeno C, Hausmann A, Schmidt S *et al.* (2022). Peering into the darkness: DNA barcoding reveals surprisingly high diversity of unknown species of Diptera (Insecta) in Germany. *Insects* **13**, 82.
- Chornelia A, Lu J, Hughes AC (2022). How to accurately delineate morphologically conserved taxa and diagnose their phenotypic disparities: species delimitation in cryptic Rhinolophidae (Chiroptera). *Frontiers in Ecology and Evolution* **10**, 854509.
- Cicero C (1996). *Sibling Species of Titmice in the Parus inornatus complex (Aves: Paridae)*. University of California Press, Oakland, California, pp. 1–217.
- Collin R, Venera-Pontón DE, Driskell AC *et al.* (2020). DNA barcoding of echinopluteus larvae uncovers cryptic diversity in neotropical echinoids. *Invertebrate Biology* **139**, e12292.
- Cong Q, Shen J, Borek D *et al.* (2017). When COI barcodes deceive: Complete genomes reveal introgression in hairstreaks. *Proceedings of the Royal Society B: Biological Sciences* **284**, 20161735.
- Cooke GM, Chao NL, Beheregaray LB (2012). Five cryptic species in the Amazonian catfish *Centromochlus eximatus* identified based on biogeographic predictions and genetic data. *PLoS ONE* **7**, e48800.
- Costa FO, DeWaard JR, Boutillier J *et al.* (2007). Biological identifications through DNA barcodes: The case of the Crustacea. *Canadian Journal of Fisheries and Aquatic Sciences* **64**, 272–95.
- Costello MJ, May RM, Stork NE (2013). Response to comments on “Can we name Earth’s species before they go extinct?”. *Science* **341**, 237.
- Damm S, Schierwater B, Hadrys H (2010). An integrative approach to species discovery in odonates: From character-based DNA barcoding to ecology. *Molecular Ecology* **19**, 3881–93.
- Darwell CT, Cook JM (2017). Cryptic diversity in a fig wasp community-morphologically differentiated species are sympatric but cryptic species are parapatric. *Molecular Ecology* **26**, 937–50.
- Dayrat B (2005). Towards integrative taxonomy. *Biological Journal of the Linnean Society* **85**, 407–17.
- de Moya RS, Brown JK, Sweet AD *et al.* (2019). Nuclear orthologs derived from whole genome sequencing indicate cryptic diversity in the *Bemisia tabaci* (Insecta: Aleyrodidae) complex of whiteflies. *Diversity* **11**, 151.
- De Queiroz K (2007). Species concepts and species delimitation. *Systematic Biology* **56**, 879–86.
- Delić T, Trontelj P, Rendoš M, Fišer C (2017). The importance of naming cryptic species and the conservation of endemic subterranean amphipods. *Scientific Reports* **7**, 3391.
- Dennis AB, Hellberg ME (2010). Ecological partitioning among parapatric cryptic species. *Molecular Ecology* **19**, 3206–25.
- Derkarabetian S, Steinmann DB, Hedin M (2010). Repeated and time-correlated morphological convergence in cave-dwelling harvestmen (Opiliones, Laniatores) from montane western North America. *PLoS ONE* **5**, e10388.
- Derycke S, De Meester N, Rigaux A *et al.* (2016). Coexisting cryptic species of the *Litoditis marina* complex (Nematoda) show differential resource use and have distinct microbiomes with high intraspecific variability. *Molecular Ecology* **25**, 2093–110.
- Després L (2019). One, two or more species? Mitonuclear discordance and species delimitation. *Molecular Ecology* **28**, 3845–47.
- Dinsdale A, Cook L, Riginos C *et al.* (2010). Refined global analysis of *Bemisia tabaci* (Hemiptera: Sternorrhyncha: Aleyrodidae: Aleyrodidae) mitochondrial

- cytochrome oxidase 1 to identify species level genetic boundaries. *Annals of the Entomological Society of America* **103**, 196–208.
- Dyer LA, Singer MS, Lill JT *et al.* (2007). Host specificity of Lepidoptera in tropical and temperate forests. *Nature* **448**, 696–99.
- Díaz S, Fargione J, Chapin FS III, Tilman D (2006). Biodiversity loss threatens human well-being. *PLoS Biology* **4**, e277.
- Ehrlich PR (1961). Has the biological species concept outlived its usefulness? *Systematic Zoology* **10**, 167–76.
- Erwin TL (1982). Tropical forests: Their richness in Coleoptera and other arthropod species. *The Coleopterists Bulletin* **36**, 74–75.
- Fan X, Yao S, Luo X *et al.* (2021). Some morphologically distinguishable hypotrich ciliates share identical 18S rRNA gene sequences—Taxonomic insights from a case study on *Oxytricha* species (Protista, Ciliophora). *Zoological Journal of the Linnean Society* **193**, 356–79.
- Felsenstein J (1985). Phylogenies and the comparative method. *The American Naturalist* **125**, 1–15.
- Fennessy J, Bidon T, Reuss F *et al.* (2016). Multi-locus analyses reveal four giraffe species instead of one. *Current Biology* **2**, 2543–49.
- Fernandez-Triana JL (2022). Turbo taxonomy approaches: Lessons from the past and recommendations for the future based on the experience with Braconidae (Hymenoptera) parasitoid wasps. *ZooKeys* **1087**, 199.
- Fišer C, Robinson CT, Malard F (2018). Cryptic species as a window into the paradigm shift of the species concept. *Molecular Ecology* **27**, 613–35.
- Funk DJ, Omland KE (2003). Species-level paraphyly and polyphyly: Frequency, causes, and consequences, with insights from animal mitochondrial DNA. *Annual Review of Ecology, Evolution, and Systematics* **34**, 397–423.
- García-Robledo C, Kuprewicz EK, Staines CL, Erwin TL, Kress WJ (2016). Limited tolerance by insects to high temperatures across tropical elevational gradients and the implications of global warming for extinction. *PNAS* **113**, 680–85.
- Ge DY, Feijó A, Abramov AV *et al.* (2021). Molecular phylogeny and morphological diversity of the *Niviventer fulvescens* species complex with emphasize on species in China. *Zoological Journal of the Linnean Society* **191**, 528–47.
- Ge DY, Lu L, Xia L *et al.* (2018). Molecular phylogeny, morphological diversity, and systematic revision of a species complex of common wild rat species in China (Rodentia, Murinae). *Journal of Mammalogy* **99**, 1350–74.
- Geml J, Laursen GA, O'NEILL K, Nusbaum HC, Taylor DL (2006). Beringian origins and cryptic speciation events in the fly agaric (*Amanita muscaria*). *Molecular Ecology* **15**, 225–39.
- Gong J, Dong J, Liu X, Massana R (2013). Extremely high copy numbers and polymorphisms of the rDNA operon estimated from single cell analysis of oligotrich and peritrich ciliates. *Protist* **164**, 369–79.
- Grundt HH, Kjølnér S, Borgen L *et al.* (2006). High biological species diversity in the arctic flora. *PNAS* **103**, 972–75.
- Han HX, Galsworthy A, Xue DY (2005). A taxonomic revision of *Limbatochlamys* Rothschild, 1894 with comments on its tribal placement in Geometrinae (Lepidoptera: Geometridae). *Zoological Studies-Taipei* **44**, 191.
- Harrison RG (1991). Molecular changes at speciation. *Annual Review of Ecology and Systematics* **22**, 281–308.
- Harrison RG (1993). Hybrids and hybrid zones: Historical perspective. In: Harrison RG, ed. *Hybrid Zones and the Evolutionary Process*. Oxford University Press, New York, pp. 3–13.
- Hartop E, Srivathsan A, Ronquist F, Meier R (2022). Towards large-scale integrative taxonomy (LIT): Resolving the data conundrum for dark taxa. *Systematic Biology* **71**, 1404–22.
- Havermans C (2016). Have we so far only seen the tip of the iceberg? Exploring species diversity and distribution of the giant amphipod *Eurythenes*. *Biodiversity* **17**, 12–25.
- Hebert PD, Cywinska A, Ball SL, DeWaard JR (2003b). Biological identifications through DNA barcodes. *Proceedings of the Royal Society B: Biological Sciences* **270**, 313–21.
- Hebert PD, Penton EH, Burns JM, Janzen DH, Hallwachs W (2004). Ten species in one, DNA barcoding reveals cryptic species in the neotropical skipper butterfly *As-traptes fulgerator*. *PNAS* **101**, 14812–17.
- Hebert PD, Ratnasingham S, De Waard JR (2003a). Barcoding animal life: Cytochrome c oxidase subunit 1 divergences among closely related species. *Proceedings of the Royal Society B: Biological Sciences* **270**, S96–S99.

- Hedin M (2015). High-stakes species delimitation in eyeless cave spiders (Cicurina, Dictynidae, Araneae) from central Texas. *Molecular Ecology* **24**, 346–61.
- Heethoff M (2018). Cryptic species—conceptual or terminological chaos? A response to Struck *et al.* *Trends in Ecology Evolution* **33**, 310.
- Hendrichs J, Vera MT, De Meyer M *et al.* (2015). Resolving cryptic species complexes of major tephritid pests. *ZooKeys* **540**, 5.
- Henry CS (1994). Singing and cryptic speciation insects. *Trends in Ecology Evolution* **9**, 388–92.
- Henry CS, Brooks SJ, Duelli P, Johnson JB (2003). A lacewing with the wanderlust: the European song species ‘Maltese’, *Chrysoperla agilis* sp.n., of the carnea group of *Chrysoperla* (Neuroptera: Chrysopidae). *Systematic Entomology* **28**, 131–48.
- Henry CS, Wells MLM (2010). Acoustic niche partitioning in two cryptic sibling species of *Chrysoperla* green lacewings that must duet before mating. *Animal Behaviour* **80**, 991–1003.
- Hinojosa JC, Koubínová D, Szenteczki MA *et al.* (2019). A mirage of cryptic species: Genomics uncover striking mitonuclear discordance in the butterfly *Thymelicus sylvestris*. *Molecular Ecology* **28**, 3857–68.
- Holland BS, Dawson MN, Crow GL, Hofman DK (2004). Global phylogeography of *Cassiopea* (Scyphozoa: Rhizostomeae): Molecular evidence for cryptic species and multiple invasions of the Hawaiian Islands. *Marine Biology* **145**, 1119–28.
- Hong D, Zhuang W, Zhu M *et al.* (2022). Positioning taxonomic research for the future. *Zoological Systematics* **47**, 185–87.
- Hou Z, Jin P, Liu H *et al.* (2022). Past climate cooling promoted global dispersal amphipods from Tian Shan montane lakes to circumboreal lakes. *Global Change Biology* **28**, 3830–45.
- Hou Z, Li S (2010). Intraspecific or interspecific variation: Delimitation of species boundaries within the genus *Gammarus* (Crustacea, Amphipoda, Gammaridae), with description of four new species. *Zoological Journal of the Linnean Society* **160**, 215–53.
- Huang J, Miao X, Wang Q *et al.* (2022). Metabarcoding reveals massive species diversity of Diptera in a subtropical ecosystem. *Ecology and Evolution* **12**, e8535.
- Hupalo K, Copilaş-Ciocianu D, Leese F, Weiss M (2023). Morphology, nuclear SNPs and mate selection reveal that COI barcoding overestimates species diversity in a Mediterranean freshwater amphipod by an order of magnitude. *Cladistics* **39**, 129–43.
- Janzen DH, Burns JM, Cong Q *et al.* (2017). Nuclear genomes distinguish cryptic species suggested by their DNA barcodes and ecology. *PNAS* **114**, 8313–18.
- Janzen DH, Hajibabaei M, Burns JM *et al.* (2005). Wedding biodiversity inventory of a large and complex Lepidoptera fauna with DNA barcoding. *Philosophical Transactions of the Royal Society B: Biological Sciences* **360**, 1835–45.
- Jiang N, Liu SX, Xue DY, Tang MJ, Xiao Q, Han HX (2014). External morphology and molecular identification of two tea Geometrid moth from southern China. *Chinese Journal of Applied Entomology* **51**, 987–1002. (In Chinese with English abstract.)
- Jiang N, Xue DY, Han HX, Cheng R (2021). Estimating hybridization as a consequence of climatic fluctuations: A case study of two geometridae species. *Molecular Phylogenetics and Evolution* **161**, 107168.
- Johnston EC, Wyatt AS, Leichter JJ, Burgess SC (2022). Niche differences in co-occurring cryptic coral species (*Pocillopora* spp.). *Coral Reefs* **41**, 767–78.
- Jones G, Barlow KE (2003). Cryptic species of echolocating bats. In: Thomas JA, Moss C, Vater M, eds. *Echolocation in Bats and Dolphins*. University of Chicago Press, Chicago, IL, pp. 345–49.
- Kaleka A, Kaur N, Bali G (2020). Larval development and molting. *Edible Insects* **560**, 17.
- Kergoat GJ, Toussaint EF, Capdevielle-Dulac C *et al.* (2015). Integrative taxonomy reveals six new species related to the Mediterranean corn stalk borer *Sesamia nonagrioides* (Lefèbvre) (Lepidoptera, Noctuidae, Sesamiina). *Zoological Journal of the Linnean Society* **175**, 244–70.
- Knowlton N (1993). Sibling species in the sea. *Annual Review of Ecology and Systematics* **24**, 189–216.
- Korshunova T, Martynov A, Bakken T, Picton B (2017). External diversity is restrained by internal conservatism: New nudibranch mollusc contributes to the cryptic species problem. *Zoologica Scripta* **46**, 683–92.
- Kress WJ, Garcia-Robledo C, Uriarte M, Erickson DL (2015). DNA barcodes for ecology, evolution, and conservation. *Trends in Ecology Evolution* **30**, 25–35.
- Kruckenhauser L, Duda M, Bartel D *et al.* (2014). Paraphyly and budding speciation in the hairy snail (Pulmonata, Hygromiidae). *Zoologica Scripta* **43**, 273–88.
- Laurance WF (2013). The race to name Earth’s species. *Science* **339**, 1275–1275.

- Lee T, Ó Foighil D (2004). Hidden Floridian biodiversity: mitochondrial and nuclear gene trees reveal four cryptic species within the scorched mussel, *Brachidontes exustus*, species complex. *Molecular Ecology* **13**, 3527–42.
- Lefébure T, Douady C J, Gouy M *et al.* (2006). Phylogeography of a subterranean amphipod reveals cryptic diversity and dynamic evolution in extreme environments. *Molecular Ecology* **15**, 1797–806.
- Lei FM, Payne RB (2002). Territorial songs of the Drongo Cuckoo complex (*Surniculus lugubris* & *S. velotinus*). *The Raffles Bulletin of Zoology* **50**, 205–13.
- Lei FM, Zhao HF, Wang G, Payne RB (2003). Vocalizations and species limits of the plaintive cuckoo (*Cacomantis merulinus*) and the brush cuckoo (*C. variolosus*). *Folia Zoology* **52**, 399–412.
- Li X, Wiens JJ (2023). Estimating global biodiversity: The role of cryptic insect species. *Systematic Biology* **72**, 391–403.
- Liu J, Li Q, Kong LF, Zheng XD (2011). Cryptic diversity in the pen shell *Atrina pectinata* (Bivalvia: Pinnidae): High divergence and hybridization revealed by molecular and morphological data. *Molecular Ecology* **20**, 4332–45.
- Liu JQ (2016). “The integrative species concept” and “species on the speciation way”. *Biodiversity science* **24**, 1004–8.
- Liu M, Liu Y, Zhang T *et al.* (2022). Integrative studies on the taxonomy and molecular phylogeny of four new *Pleuronema* species (Protozoa, Ciliophora, Scuticociliatia). *Marine Life Science Technology* **4**, 179–200.
- Liu Y, Song Y, Li XY (2010). Application of DNA barcoding technology based on mitochondrial COI gene in insect molecular identification. *Plant Quarantine* **24**, 46–50. (In Chinese with English abstract.)
- Luo C, Yao Y, Wang RJ, Yan FM, Hu DX, Zhang ZL (2002). The use of mitochondrial cytochrome oxidase I (Mt COI) gene sequences for the identification of biotypes of *Bemisia tabaci* (Gennadius) in China. *Acta Entomologica Sinica* **45**, 759–63. (In Chinese with English abstract.)
- Mallet J (2005). Hybridization as an invasion of the genome. *Trends in Ecology and Evolution* **20**, 229–37.
- Marshall DC, Cooley JR (2000). Reproductive character displacement and speciation in periodical cicadas, with description of a new species, 13-year Magicicada *neotredecim*. *Evolution* **54**, 1313–25.
- Martinet B, Lecocq T, Brasero N *et al.* (2019). Integrative taxonomy of an arctic bumblebee species complex highlights a new cryptic species (Apidae: *Bombus*). *Zoological Journal of the Linnean Society* **187**, 599–621.
- Mata VA, Ferreira S, Campos RM *et al.* (2021). Efficient assessment of nocturnal flying insect communities by combining automatic light traps and DNA metabarcoding. *Environmental DNA* **3**, 398–408.
- Mayr AV, Keller A, Peters MK *et al.* (2021). Cryptic species and hidden ecological interactions of halictine bees along an elevational gradient. *Ecology and Evolution* **11**, 7700–7712.
- Mayr E (1963). *Animal Species and Evolution*. The Belknap Press, Cambridge, MA.
- Mayr E (1982). Of what use are subspecies? *The Auk* **99**, 593–95.
- Meier R, Blaimer BB, Buenaventura E *et al.* (2022). A reanalysis of the data in Sharkey *et al.*’s 2021 minimalist revision reveals that BINs do not deserve names, but BOLD Systems needs a stronger commitment to open science. *Cladistics* **38**, 264–75.
- Michener CD (2007). *The Bees of the World*. Johns Hopkins University Press, Baltimore, MD, USA.
- Moen DS, Irschick DJ, Wiens JJ (2013). Evolutionary conservatism and convergence both lead to striking similarity in ecology, morphology and performance across continents in frogs. *Proceedings of the Royal Society B: Biological Sciences* **280**, 20132156.
- Mora C, Tittensor DP, Adl S *et al.* (2011). How many species are there on Earth and Murray in the ocean? *PLoS Biology* **9**, e1001127.
- Ge D, Feijó A, Wen Z *et al.* (2022). Ancient introgression underlying the unusual mito-nuclear discordance and coat phenotypic variation in the Mouping pika. *Diversity and Distributions* **28**, 2593–609.
- Murray TE, Fitzpatrick U, Brown MJ, Paxton RJ (2008). Cryptic species diversity in a widespread bumble bee complex revealed using mitochondrial DNA RFLPs. *Conservation Genetics* **9**, 653–66.
- Nair A, Gopalan S V, George S *et al.* (2012). High cryptic diversity of endemic Indirana frogs in the Western Ghats biodiversity hotspot. *Animal Conservation* **15**, 489–98.
- Narins PM (1983). Divergence of acoustic communication systems of two sibling species of eleutherodactylid frogs. *Copeia* **4**, 1089–90.
- Nevo E (2001). Evolution of genome-phenome diversity under environmental stress. *PNAS* **98**, 6233–40.
- Niemiller ML, Graening GO, Fenolio DB *et al.* (2013). Doomed before they are described? The need for

- conservation assessments of cryptic species complexes using an amblyopsid cavefish (Amblyopsidae: *Typhlichthys*) as a case study. *Biodiversity and Conservation* **22**, 1799–820.
- Orr MC, Ascher JS, Bai M, Chesters D, Zhu CD (2020). Three questions: How can taxonomists survive and thrive worldwide? *Megataxa* **1**, 19–27.
- Orr MC, Feijó A, Chesters D, Vogler AP, Bossert S *et al.* (2022). Six steps for building a technological knowledge base for future taxonomic work. *National Science Review* **9**, nwac284.
- Ottenburghs J (2020). Ghost introgression: Spooky gene flow in the distant past. *Bioessays* **42**, e2000012.
- Padial JM, Miralles A, De la Riva I, Vences M (2010). The integrative future of taxonomy. *Frontiers in Zoology* **7**, 16.
- Pearce T (2011). Convergence and parallelism in evolution: A neo-Gouldian account. *The British Journal for the Philosophy of Science* **63**, 429–48.
- Pekár S (2014). Comparative analysis of passive defences in spiders (Araneae). *Journal of Animal Ecology* **83**, 779–90.
- Peng JL, Wang XZ, He SP (2008). The progress and application of DNA barcoding. *Acta Hydrobiologica Sinica* **32**, 916–19. (In Chinese with English abstract.)
- Pfenninger M, Schwenk K (2007). Cryptic animal species are homogeneously distributed among taxa and biogeographical regions. *BMC Evolutionary Biology* **7**, 121.
- Pfingstl T, Lienhard A, Baumann J, Koblmüller S (2021). A taxonomist's nightmare—Cryptic diversity in Caribbean intertidal arthropods (Arachnida, Acari, Oribatida). *Molecular Phylogenetics and Evolution* **163**, 107240.
- Ritter CD, Häggqvist S, Karlsson D *et al.* (2019). Biodiversity assessments in the 21st century: The potential of insect traps to complement environmental samples for estimating eukaryotic and prokaryotic diversity using high-throughput DNA metabarcoding. *Genome* **62**, 147–59.
- Rothschild LJ, Mancinelli RL (2001). Life in extreme environments. *Nature* **409**, 1092–101.
- Rowe KC, Aplin KP, Baverstock PR, Moritz C (2011). Recent and rapid speciation with limited morphological disparity in the genus *Rattus*. *Systematic Biology* **60**, 188–203.
- Ruppert KM, Kline RJ, Rahman MS (2019). Past, present, and future perspectives of environmental DNA (eDNA) metabarcoding: A systematic review in methods, monitoring, and applications of global eDNA. *Global Ecology and Conservation* **17**, e00547.
- Ruxton GD (2009). Non-visual crypsis: A review of the empirical evidence for camouflage to senses other than vision. *Philosophical Transactions of the Royal Society B: Biological Sciences* **364**, 549–57.
- Schönrogge K, Barr B, Wardlaw JC *et al.* (2002). When rare species become endangered: cryptic speciation in myrmecophilous hoverflies. *Biological Journal of the Linnean Society* **75**, 291–300.
- Scriven JJ, Whitehorn PR, Goulson D, Tinsley MC (2016). Niche partitioning in a sympatric cryptic species complex. *Ecology and Evolution* **6**, 1328–39.
- Shao C, Hu C, Fan Y, Warren A, Lin X (2019). Morphology, morphogenesis and molecular phylogeny of a fresh water ciliate, *Monomicrocaryon euglenivorum euglenivorum* (Ciliophora, Oxytrichidae). *European Journal of Protistology* **68**, 25–36.
- Sharkey MJ, Janzen DH, Hallwachs W *et al.* (2021). Minimalist revision and description of 403 new species in 11 subfamilies of Costa Rican braconid parasitoid wasps, including host records for 219 species. *ZooKeys* **1013**, 1–665.
- Shen Y, Hubert N, Huang Y *et al.* (2019). DNA barcoding the ichthyofauna of the Yangtze River: Insights from the molecular inventory of a mega-diverse temperate fauna. *Molecular Ecology Resources* **19**, 1278–91.
- Simmons RB, Scheffer SJ (2004). Evidence of cryptic species within the pest *Copitarsia decolora* (Guenée) (Lepidoptera: Noctuidae). *Annals of the Entomological Society of America* **97**, 675–80.
- Sokal RR, Crovello TJ (1970). The biological species concept: A critical evaluation. *The American Naturalist* **104**, 127–53.
- Stamos DN (2003). *The Species Problem: Biological Species, Ontology, and the Metaphysics of Biology*. Lexington Books, Lanham, MA.
- Steane DA, Potts BM, McLean E *et al.* (2015). Genome-wide scans reveal cryptic population structure in a dry-adapted eucalypt. *Tree Genetics & Genomes* **11**, 33.
- Stork NE (2018). How many species of insects and other terrestrial arthropods are there on Earth? *Annual Review of Entomology* **63**, 31–45.
- Struck TH, Cerca J (2019). Cryptic species and their evolutionary significance. In: *Encyclopedia of Life Sciences*. Wiley, New York, pp. 1–9. <https://doi.org/10.1002/9780470015902.a0028292>

- Struck TH, Feder JL, Bendiksy M *et al.* (2018). Finding evolutionary processes hidden in cryptic species. *Trends in Ecology Evolution* **33**, 153–63.
- Suh KI, Hwang JM, Bae YJ, Kang JH (2019). Comprehensive DNA barcodes for species identification and discovery of cryptic diversity in mayfly larvae from South Korea: Implications for freshwater ecosystem biomonitoring. *Entomological Research* **49**, 46–54.
- Swift HF, Daglio LG, Dawson MN (2016). Three routes to cryptic species: stasis, convergence, and Sun parallelism in the *Mastigias* species complex (Scyphozoa, Rhizostomeae). *Molecular Phylogenetics and Evolution* **99**, 103–15.
- Tang M, Hardman CJ, Ji Y *et al.* (2015). High-throughput monitoring of wild bee diversity and abundance via mitogenomics. *Methods in Ecology and Evolution* **6**, 1034–43.
- Tattersall I (2007). Madagascar's lemurs: cryptic diversity or taxonomic inflation? *Evolutionary Anthropology* **16**, 12–23.
- Tautz D, Arctander P, Minelli A, Thomas RH, Vogler AP (2002). DNA points the way ahead in taxonomy. *Nature* **418**, 479.
- Tautz D, Arctander P, Minelli A, Thomas RH, Vogler AP (2003). A plea for DNA taxonomy. *Trends in Ecology & Evolution* **18**, 70–74.
- Trontelj P, Douady CJ, Fišer C *et al.* (2009). A molecular test for cryptic diversity in ground water: How large are the ranges of macro-stygobionts? *Freshwater Biology* **54**, 727–44.
- Tyagi K, Kumar V, Singha D *et al.* (2017). DNA Barcoding studies on Thrips in India: Cryptic species and species complexes. *Scientific Reports* **7**, 4898.
- Utsunomiya YT, Pérez O'Brien AM, Sonstegard TS *et al.* (2013). Detecting loci under recent positive selection in dairy and beef cattle by combining different genome-wide scan methods. *PLoS ONE* **8**, e64280.
- Van Campenhout J, Derycke S, Moens T, Vanreusel A (2014). Differences in life-histories refute ecological equivalence of cryptic species and provide clues to the origin of bathyal *Halomonhystera* (Nematoda). *PLoS ONE* **9**, e111889.
- Vodá R, Dapporto L, Dincă V, Vila R (2015a). Cryptic matters: Overlooked species generate most butterfly beta-diversity. *Ecography* **38**, 405–9.
- Vodá R, Dapporto L, Dincă V, Vila R (2015b). Why do cryptic species tend not to co-occur? A case study on two cryptic pairs of butterflies. *PLoS ONE* **10**, e0117802.
- Vrijenhoek RC, Schutz SJ, Gustafson RG *et al.* (1994). Cryptic species of deep-sea clams (Mollusca: Bivalvia: Vesicomidae) from hydrothermal vent and cold-water seep environments. *Deep Sea Research Part I: Oceanographic Research Papers* **41**, 1171–89.
- Wang T, Zhang YP, Yang ZY, Liu Z, Du YY (2020). DNA barcoding reveals cryptic diversity in the underestimated genus *Triplophysa* (Cypriniformes: Cobitidae, Nemacheilinae) from the northeastern Qinghai-Tibet Plateau. *BMC Evolutionary Biology* **20**, 151.
- Wang W, Dai C, Alström P *et al.* (2014). Past hybridization between two East Asian long-tailed tits (*Aegithalos bonvaloti* and *A. fuliginosus*). *Frontiers in Zoology* **11**, 40.
- Weigand H, Weiss M, Cai H *et al.* (2017). Deciphering the origin of mito-nuclear discordance in two sibling caddisfly species. *Molecular Ecology* **26**, 5705–15.
- Weisbecker V, Archer M (2008). Parallel evolution of hand anatomy in kangaroos and vombatiform marsupials: Functional and evolutionary implications. *Palaeontology* **51**, 321–38.
- Westwood JO (1833). On the probable number of species of insects in the creation; together with descriptions of several minute Hymenoptera. *The Magazine of Natural History and Journal of Zoology, Botany, Mineralogy, Geology, and Meteorology* **6**, 116–123.
- Wilkins JS (2006). The concept and causes of microbial species. *History and Philosophy of the Life Sciences* **28**, 389–407.
- Wilkins JS (2009). *Defining Species: A Sourcebook from Antiquity to Today*, vol. 203. Peter Lang, New York.
- Williams PH, Brown MJ, Carolan JC *et al.* (2012). Unveiling cryptic species of the bumblebee subgenus *Bombus s. str.* worldwide with COI barcodes (Hymenoptera: Apidae). *Systematics and Biodiversity* **10**, 21–56.
- Willig MR, Kaufman DM, Stevens RD (2003). Latitudinal gradients of biodiversity: Pattern, process, scale, and synthesis. *Annual Review of Ecology, Evolution, and Systematics* **34**, 273–309.
- Winker K (2005). Sibling species were first recognized by William Derham (1718). *Auk* **122**, 706–7.
- Witt JDS, Threlloff DL, Hebert PDN (2006). DNA barcoding reveals extraordinary cryptic diversity in an amphipod genus: Implications for desert spring conservation. *Molecular Ecology* **15**, 3073–82.
- Wu HC, Lin RC, Hung HY *et al.* (2011). Molecular and morphological evidences revealed the cryptic species in the vinaceous rosefinch *Carpodacus*

- vinaceus* (Fringillidae; Aves). *Zoologica Scripta* **40**, 468–78.
- Xiao JH, Xiao H, Huang DW (2004). DNA barcoding: New approach of biological taxonomy. *Acta Zoologica Sinica* **50**, 852–55. (In Chinese with English abstract.)
- Xiong F, Shu L, Gan X, Zeng H, He S, Peng Z (2022). Methodology for fish biodiversity monitoring with environmental DNA metabarcoding: the primers, databases and bioinformatic pipelines. *Water Biology and Security* **1**, 100007.
- Xu W, Che J (2019). From cryptic species to biodiversity conservation in China: Status and prospects. *Scientia Sinica Vitae* **49**, 519–30. (In Chinese with English abstract.)
- Yamamoto S, Sota T (2007). Phylogeny of the Geometridae and the evolution of winter moths inferred from a simultaneous analysis of mitochondrial and nuclear genes. *Molecular Phylogenetics and Evolution* **44**, 711–23.
- Yan F, Lü J, Zhang B *et al.* (2018). The Chinese giant salamander exemplifies the hidden extinction of cryptic species. *Current Biology* **28**, R590–R592.
- Yang XJ, Lei FM (2008). Song structure and its microgeographic variation in the Chinese bulbul *Pycnonotus sinensis*. *Acta Zoologica Sinica* **54**, 630–639. (In Chinese with English abstract.)
- Yu HJ, Wang Q (2021). Application of DNA barcoding for species identification in aquatic animal. *Anhui Agricultural Sciences* **49**, 1–3. (In Chinese with English abstract.)
- Zakharov EV, Lobo NF, Nowak C, Hellmann JJ (2009). Introgression as a likely cause of mtDNA paralogy in two allopatric skippers (Lepidoptera: Hesperidae). *Heredity (Edinb)* **102**, 590–99.
- Zenker MM, Specht A, Fonseca VG (2020). Assessing insect biodiversity with automatic light traps in Brazil: Pearls and pitfalls of metabarcoding samples in preservative ethanol. *Ecology and Evolution* **10**, 2352–66.
- Zhang DY, Lin K, Hanski I (2004). Coexistence of cryptic species. *Ecology Letters* **7**, 165–69.
- Zhang Y, Li S (2014). A spider species complex revealed high cryptic diversity in South China caves. *Molecular Phylogenetics and Evolution* **79**, 353–58.
- Zhou QS, Polaszek A, Qin YG *et al.* (2017). Parasitoid–host associations of the genus *Coccophagus* (Hymenoptera: Aphelinidae) in China. *Zoological Journal of the Linnean Society* **182**, 38–49.
- Zhou X, Guang X, Sun D *et al.* (2018). Population genomics of finless porpoises reveal an incipient cetacean species adapted to freshwater. *Nature Communications* **9**, 1276.
- Zhou X, Li Y, Liu S *et al.* (2013). Ultra-deep sequencing enables high-fidelity recovery of biodiversity for bulk arthropod samples without PCR amplification. *Giga-Science* **2**, 4.
- Zhu C, Luo A, Bai M *et al.* (2022). A joint call for actions to advance taxonomy in China. *Zoological Systematics* **47**, 188–97.

Cite this article as:

Cheng R, Luo A, Orr M *et al.* (2025). Cryptic diversity begets challenges and opportunities in biodiversity research. *Integrative Zoology* **20**, 33–49. <https://doi.org/10.1111/1749-4877.12809>